A INTER DEPARTMENTAL PROJECT REPORT ON

A COMPARATIVE ANALYSIS OF MACHINE LEARNING AND DEEP LEARNING MODELS FOR TIME-SERIES PREDICTION OF COEFFICIENT OF FRICTION

Submitted in partial fulfilment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

in

MECHANICAL ENGINEERING

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CERTIFICATE

This is to certify that the inter departmental project entitled **"A COMPARATIVE ANALYSIS OF MACHINE LEARNING AND DEEP LEARNING MODELS FOR TIME-SERIES PREDICTION OF COEFFICIENT OF FRICTION"** being submitted by([G.Sandeep] (231FA08004)), ([M.Banu Teja] & [231FA08017]), ([P.Kartheek] & [231FA08026]) in partial fulfilment of Bachelor of Technology in the Department of Mechanical Engineering, Vignan's Foundation For Science Technology & Research (Deemed to be University), Vadlamudi, Guntur District, Andhra Pradesh, India, is a bonafide work carried out by them under my guidance and supervision.

Head of the Department — (Dr. N.Narayanarao)

Guide — (Dr.P.Satish)

DECLARATION

We hereby declare that our project work described in the interdepartmental project titled **"A COMPARATIVE ANALYSIS OF MACHINE LEARNING AND DEEP LEARNING MODELS FOR TIME-SERIES PREDICTION OF COEFFICIENT OF FRICTION"** which is being submitted by us for the partial fulfilment in the department of MECHANICAL ENGINEERING, Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur, Andhra Pradesh, and the result of investigations are carried out by us under the guidance of Dr.Perabathula Satish .

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ABSTRACT

Predicting the Coefficient of Friction (COF) is a critical aspect of tribological analysis and material science. Traditional experimental methods for determining COF, especially over long durations, are often time-consuming and resource-intensive. This project investigates the application of machine learning (ML) models for accurately forecasting the time-series behaviour of COF. A comprehensive comparative analysis was conducted using five distinct ML and Deep Learning algorithms: Random Forest, AdaBoost, XGBoost, CatBoost, and Long Short-Term Memory (LSTM). These models were trained and tested on experimental data, using 'Time' as the sole input feature to predict the corresponding 'COF'. The performance of each model was rigorously evaluated using standard regression metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R-squared (R^2). The results demonstrated that while all models achieved high accuracy, the Random Forest algorithm provided the superior performance. It achieved the lowest MAE (0.003725), the lowest RMSE (0.00821), and the highest R^2 score (0.991), outperforming the advanced gradient boosting and deep learning models for this dataset. This study confirms that machine learning, and specifically the Random Forest model, is an exceptionally effective and robust tool for forecasting the complex, time-dependent behavior of friction, presenting a viable and efficient alternative to prolonged experimental testing.

OBJECTIVES

1. To extract and digitize time-series tribological data (Time vs. COF) from a published research paper.
2. To implement and train a suite of machine learning models (Random Forest, AdaBoost, XGBoost, CatBoost) and a deep learning model (LSTM) on the acquired dataset.
3. To rigorously evaluate the predictive performance of each model using standard metrics (MAE, MSE, RMSE, and R^2).
4. To compare the models and identify the most accurate and robust algorithm for forecasting the Coefficient of Friction based on the paper's data.
5. To analyze the feasibility of using the best model as an efficient alternative to replicating the original paper's experimental testing.

CHAPTER 1

INTRODUCTION

Tribology, the science and engineering of interacting surfaces in relative motion, is a cornerstone of modern mechanical engineering. It encompasses the study of friction, wear, and lubrication—three phenomena that are present in virtually every mechanical system, from microscopic MEMS devices to massive power-generation turbines. Of these, friction is arguably the most fundamental. It is the resisting force encountered when one body moves or attempts to move over another. This force, which originates from complex electromagnetic interactions between molecules at an interface, has profound implications for the efficiency, durability, and performance of all moving machinery.

Historically, the study of friction has been a double-edged sword, as the force itself is both a critical enabler and a persistent obstacle to technological progress. The immense negative impact of friction is the most common focus of engineering. A significant portion of the world's total energy consumption is expended simply to overcome frictional forces in machines and transportation. This energy loss does not simply vanish; it primarily converts into heat. This waste heat is a major problem, as it can degrade lubricant properties, alter material tolerances through thermal expansion, and lead to catastrophic component failure or system seizure.

Beyond energy loss, friction is the direct mechanism responsible for wear. Wear, the progressive loss of material from a surface due to mechanical action, is one of the most frequent causes of component breakdown. As surfaces slide against each other, friction causes material to be sheared, abraded, and fatigued. This leads to component failure, reduced service life, and the need for costly maintenance and replacement. The economic impact is staggering, with estimates placing the cost of friction and wear at a significant percentage of the Gross National Product for any industrialized nation. Thus, the engineering drive to minimize friction and wear is a primary motivation for tribological research, aimed at creating systems that are more efficient, more durable, and more reliable.

While friction is often cast as an antagonist, it is also a force of indispensable utility. Without friction, the world as we know it would be impossible. Simple actions like walking, writing, or grasping an object would be unachievable. In engineering, controlled friction is the entire principle behind essential systems. **Vehicle brakes**, for example, are designed to maximize friction, converting kinetic energy into heat to stop a vehicle safely. Similarly, **clutches** rely on friction to transmit power from an engine to a transmission. Even static fasteners, such as nuts and bolts, rely on friction to hold assemblies together against vibration and external loads.

Therefore, the ultimate goal of the tribologist is not to *eliminate* friction, but to *control* it. To do this, the force must be quantified. The primary parameter used for this quantification is the **Coefficient of Friction (COF)**, typically denoted by the Greek letter μ (mu). The COF is a dimensionless scalar value defined as the ratio of the friction force to the normal force pressing the two surfaces together.

However, the COF is not a simple, intrinsic material property like density or thermal conductivity. It is a complex **system property** that depends on a multitude of variables. These include the properties of both materials, the surface roughness, the applied load, the relative velocity, and the ambient temperature. Furthermore, the COF is not a static value; it is a **dynamic, time-dependent** parameter. In nearly all mechanical systems, the COF changes dramatically during the initial "running-in" or "wear-in" period. During this phase, surface asperities are flattened, oxide layers are scraped away, and the true contact area evolves. Only after this volatile initial phase does the system typically settle into a more stable "steady-state" friction regime.

This complex, non-linear, and multi-variable nature of friction makes it extraordinarily difficult to predict using first principles or simple analytical models. While classical laws provide a basic approximation, they fail to capture the dynamic, real-world behaviour of tribological systems.

Given the immense complexity of tribological systems, traditional analytical and physics-based models are often insufficient. These models struggle to account for the countless interacting variables and the non-linear, time-dependent nature of friction and wear. This challenge has opened the door for a powerful new approach: data-driven modeling. The rise of computational power has led to the emergence of "Triboinformatics," a field that leverages machine learning (ML) and artificial intelligence (AI) to analyze, model, and predict tribological behaviour.

Machine learning, a subset of AI, is a "rising star" in tribology because it excels at solving the very problems that traditional methods cannot. Instead of relying on a pre-defined physical equation, ML algorithms can learn complex patterns and non-linear relationships directly from large datasets. Tribology is field rich in experimental data; ML provides the tools to "support sorting through the complexity of patterns and identifying trends within the multiple interacting features".

This data-driven approach is now being applied across all facets of tribology, including composite materials, surface engineering, and lubricant development. A review of the literature by Marian & Tremmel (2021) shows that a wide variety of ML algorithms are being successfully implemented. The most common among these are Artificial Neural Networks (ANNs), Decision Trees, Random Forest (RF), and Support Vector Machines (SVMs).

These computational tools are primarily used for the express purpose of "predicting and optimizing" the tribological behavior of systems. Instead of running hundreds of costly and time-consuming physical experiments, engineers can now use ML models trained on a smaller dataset to predict outcomes. This accelerates the design of new materials and components, allowing for virtual screening of "what-if" scenarios. For example, a model can predict the wear rate of a new composite material before it is even fabricated or forecast the friction behavior of a bearing under different load conditions.

The practical application of machine learning in tribological research is well-documented in recent literature. For instance, Sivaraman et al. (2025) conducted a comprehensive study to predict the wear rate and phase formation of High Entropy Alloy (HEA) coatings. They employed a suite of ten different ML models, including AdaBoost, XGBoost, and CatBoost. Their results demonstrated the high effectiveness of these algorithms, with XGBoost achieving the highest predictive accuracy for wear rate (an R^2 of 0.98) Their work also used Random Forest for the classification task of predicting material phases, achieving an accuracy of 98.5%. This strongly supports the use of these specific ensemble models for complex materials prediction.

This project focuses on one of the most challenging tasks in tribology: the time-series prediction of COF. This specific problem was addressed by Kolev & Drenchev (2023), who are the source of the data for this report. Their paper presented experimental data on the "coefficient of friction... as a function of time" for AlSi10Mg-Al2O3 composites. To model this behaviour, they specifically employed the Extreme Gradient Boosting (XGBoost) model. Their results were excellent, achieving R^2 values as high as 0.9971 on their test set, proving that XGBoost is an exceptionally powerful tool for this task.

However, the study by Kolev & Drenchev focused *only* on the performance of XGBoost. While other studies have compared models for predicting static wear rates, a direct, head-to-head comparison of different leading-edge models for this specific, univariate time-series COF problem has not been thoroughly explored.

This project aims to fill that gap. Using the same dataset published by Kolev & Drenchev, this report will conduct a comprehensive comparative analysis. We will implement not only the model from the original paper (XGBoost) but also four other powerful algorithms: Random Forest, AdaBoost, CatBoost, and the deep learning model Long Short-Term Memory (LSTM). The objective is to rigorously evaluate and compare these five models to determine which provides the most accurate, reliable, and robust forecast for the dynamic, time-dependent behavior of the Coefficient of Friction.

CHAPTER 2

LITERATURE REVIEW

2.1. The Rise of "Triboinformatics"

Tribology, the study of friction, wear, and lubrication, is an inherently complex field. The performance of a tribological system depends on a multitude of interacting features and processes, making it difficult to fully describe with traditional mathematical or differential equations. In recent years, Machine Learning (ML) and Artificial Intelligence (AI) have emerged as "rising stars" in this discipline, offering powerful new methods to identify trends and sort through this complexity.

This integration of data science with tribology has given rise to the new field of "Triboinformatics". The primary advantage of ML is its ability to handle high-dimensional problems and datasets, allowing for the identification of relevant relationships and causality from existing data. A quantitative survey of the field by Marian & Tremmel (2021) revealed that the vast majority of research (76%) is built upon experimental data. The goals of this research are manifold, but the most common intention (53% of studies) is for "predicting and optimizing" the tribological behavior of systems.

This data-driven approach is now being applied across all facets of tribology, including composite materials, surface engineering, drive technology, and lubricant formulation.

2.2. A Survey of Deployed Machine Learning Models

The literature shows a wide variety of ML algorithms being successfully used to solve tribological problems. The survey by Marian & Tremmel found that Artificial Neural Networks (ANNs) are the most common, appearing in roughly three-quarters of the published work in the field.

Domination of Artificial Neural Networks (ANNs):

ANNs have been extensively used, particularly in the study of composite materials.

- Padhi & Satapathy (2013) applied an ANN to train a multi-layered network to predict the tribological behaviour of epoxy composites reinforced with short glass fibers.
- Egala et al. (2021) also investigated epoxy composites, testing 73 different ANN architectures to find the best prediction model for wear, temperature, and COF.
- This trend extends to thermoplastic composites, where Velten et al. (2000) evaluated ANNs for predicting properties of short-fiber thermoplastic (PA) composites.
- Gyurova et al. (2010) modeled the sliding wear and friction of polyphenylene sulfide (PPS) composites using ANNs, noting that the prediction quality was highly dependent on the size of the dataset, with larger datasets yielding significantly lower errors.

While ANNs are prevalent, they are not the only, or always the best, solution. The choice of the ML approach is highly dependent on the specific problem.

The Shift to Comparative Analysis and Ensemble Models:

More recent and comprehensive studies have moved beyond using a single model and instead focus on a comparative analysis of multiple algorithms to find the most robust predictor. This methodology is central to modern triboinformatics.

- A key example cited by Marian & Tremmel is the work of Hasan et al. (2021), who compared five different ML techniques—ANN, K-Nearest Neighbour (KNN), Support Vector Machine (SVM), Gradient Boosting Machine (GBM), and Random Forest (RF)—to predict the friction and wear of aluminium-graphite composites. Their findings were significant: Random Forest outperformed the other algorithms in predicting wear, while GBM (a boosting algorithm like XGBoost) had the highest accuracy for predicting friction. This strongly suggests that ensemble methods (RF and GBM) are top-tier candidates for tribological prediction.
- This comparative approach is exemplified by the work of Sivaraman et al. (2025), who investigated the prediction of wear rate and phase formation in High Entropy Alloy (HEA) coatings. Their problem was similar to many in tribology: accessing wear behavior is "complex and time-consuming".

- **Methodology:** They sourced their data from 80 peer-reviewed literature samples and split it 80:20 for training and testing.
- **Wear Rate Prediction (Regression):** They compared ten different ML models, including AdaBoost, XGBoost, CatBoost, Random Forest (via GBRT), and SVM. Their results showed that the boosting algorithms were exceptionally effective, achieving an average R^2 of 0.96. The top-performing model was XGBoost, which achieved the highest predictive effectiveness with an R^2 of 0.98. The SVM-RBF model was the second-best, with an R^2 of 0.97.
- **Phase Prediction (Classification):** In a separate task, they compared four classification models (RF, GNB, ANN, Logistic Regression). Here, the Random Forest model was the clear winner, exhibiting the best accuracy at 98.5%.

The work of Sivaraman et al. provides a powerful template for this project, as it validates the methodology of (1) sourcing data from literature, (2) comparing a suite of ensemble models, and (3) it confirms the high performance of XGBoost, CatBoost, AdaBoost, and Random Forest for tribological prediction.

2.3. The Specific Task: Time-Series COF Prediction

While many studies like Sivaraman's focus on predicting a single, static value (like final wear rate), a more complex challenge is predicting the dynamic, time-dependent behavior of friction as it evolves. This project focuses specifically on this task: predicting the Coefficient of Friction as a function of Time.

This exact problem was the subject of the paper by Kolev & Drenchev (2023), which serves as the direct precedent and data source for this project.

- **Objective:** Their data article presented experimental data on the wear behavior and COF "as a function of time" for open-cell AlSi10Mg-Al2O3 composites.
- **Methodology:** The data was collected from a pin-on-disk tribometer with a sliding time of 420 seconds. The authors then used this time-series data to train a machine learning model.

- **Model Selection:** Kolev & Drenchev employed one specific ML model for their prediction: Extreme Gradient Boosting (XGBoost). They used an 80:20 train-test split and 5-fold cross-validation.
- **Results:** The XGBoost model proved to be exceptionally accurate for this time-series forecasting task. The authors reported performance metrics for their "test-set" and "validation-set," which included R^2 values as high as **0.9971** and **0.9944**. These results strongly validate the use of XGBoost for predicting COF over time.

2.4. The Research Gap

The literature provides a clear and logical path forward.

1. **Marian & Tremmel (2021)** establish that ML is a primary tool for "predicting and optimizing" tribological systems, and that comparative studies (like Hasan et al.) show Random Forest and Gradient Boosting are top performers.
2. **Sivaraman et al. (2025)** provide a robust methodology for comparing a large suite of ensemble models (XGBoost, CatBoost, AdaBoost, RF) and prove that **XGBoost** is a best-in-class model for regression tasks in materials science.
3. **Kolev & Drenchev (2023)** *specifically* validate the use of XGBoost for the *exact problem* this project is undertaking: predicting time-series COF data for aluminum composites.

However, the Kolev & Drenchev study, while successful, limited its analysis to *only* the XGBoost model. The Sivaraman study, while comprehensive, focused on predicting a single *wear rate* value, not a time-series curve.

Therefore, the clear research gap is the lack of a comprehensive, direct comparison of multiple leading-edge models for the specific task of time-series COF prediction on this dataset. This project will fill this gap by applying the comparative methodology of Sivaraman et al. to the specific time-series problem and dataset presented by Kolev & Drenchev. We will test Random Forest, AdaBoost, CatBoost, and the deep learning model LSTM against the established XGBoost benchmark to determine which algorithm truly provides the most accurate and robust forecast.

CHAPTER 3

METHODOLOGY

The project follows a structured data-driven workflow, from data collection to model evaluation. The entire process was conducted using the Python programming language, leveraging libraries such as Scikit-Learn, TensorFlow/Keras, XGBoost, and CatBoost.

3.1. Data Acquisition and Preprocessing:

- Dataset: The primary dataset was acquired from the research paper "Data on the coefficient of friction and its prediction by a machine learning model as a function of time for open-cell AlSi10Mg-Al_2O_3 composites..." by Kolev & Drenchev[1]. The time-series data (Time vs. COF) was digitized from the published graphs to create a usable dataset.
- Train-Test Split: The dataset was split chronologically into a training set and a testing set. Following established machine learning practice, 80% of the data was used for training the models, and the remaining 20% was reserved for testing and evaluation.

3.2. Model Implementation:

Five state-of-the-art models were implemented to conduct a comprehensive comparative analysis:

Ensemble Models (Machine Learning):

3.2.1. Random Forest (RF)

Random Forest(Fig-1) is an ensemble learning method that operates by constructing a multitude of decision trees during training. For regression tasks like COF prediction, it outputs the mean prediction of the individual trees. It corrects for decision trees' habit of overfitting to their training set, providing a robust baseline for our comparison.

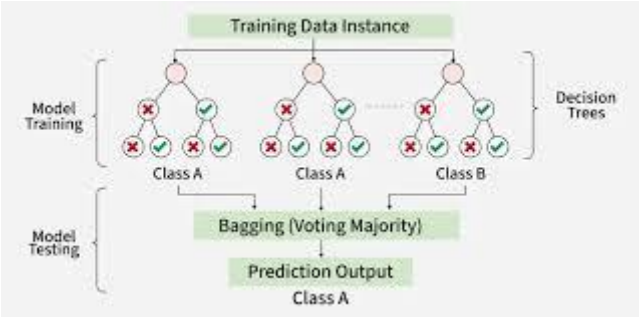


Fig-1

3.2.2 AdaBoost (Adaptive Boosting)

AdaBoost (Fig-2)works by fitting a sequence of weak learners (typically small decision trees) on repeatedly modified versions of the data. The predictions from all of them are then combined through a weighted majority vote (or sum) to produce the final prediction. It focuses more on difficult-to-predict data points in subsequent iterations, potentially capturing sharp changes in friction better than simple averaging.

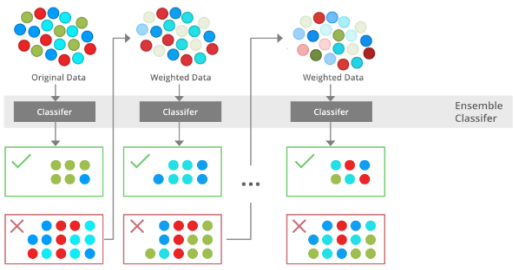


Fig-2

3.2.3 XGBoost (Extreme Gradient Boosting)

XGBoost(Fig-3) is an optimized distributed gradient boosting library designed to be highly efficient, flexible, and portable. It implements machine learning algorithms under the Gradient Boosting framework. It provides a parallel tree boosting that solves many data science problems in a fast and accurate way. Its regularization techniques help in preventing overfitting when modeling the noisy friction data.

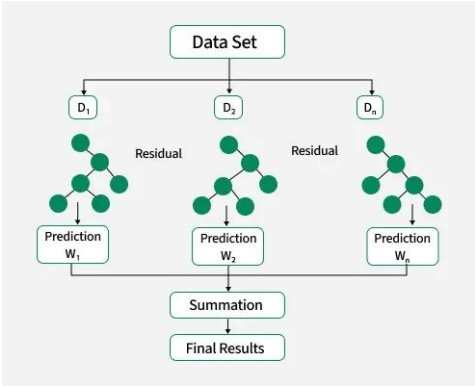


Fig-3

3.2.4 CatBoost (Categorical Boosting)

CatBoost(Fig-4) is another high-performance open-source library for gradient boosting on decision trees. It is known for its superior handling of categorical data (though less relevant here if only using time) and its robust default parameters that often require less tuning than XGBoost. It uses ordered boosting to fight prediction shift, which can be beneficial for time-series data.

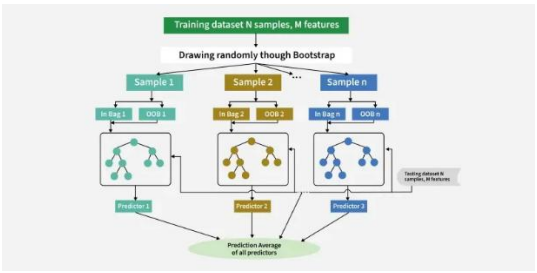


Fig-4

Deep Learning Model:

3.2.5 LSTM (Long Short-Term Memory):

LSTM is a special kind of Recurrent Neural Network (RNN), capable of learning long-term dependencies. Unlike standard feedforward neural networks, LSTMs have feedback connections, allowing them to process entire sequences of data (like our time-series). They are explicitly designed to avoid the long-term

dependency problem, making them theoretically ideal for remembering past friction states to predict future ones.

3.3. Evaluation Metrics:

To objectively compare the performance of each model, four standard regression metrics were used, which are widely accepted in machine learning research4444:

3.3.1. Mean Absolute Error (MAE): Measures the average magnitude of the errors in a set of predictions, without considering their direction.

3.3.2. Root Mean Squared Error (RMSE): The square root of the average of squared differences between prediction and actual observation. It gives higher weight to large errors.

$$MSE=\frac{1}{n}\sum_{i=1}^{n}(y_{actual}-y_{predicted})^2$$

$$RMSE=\sqrt{MSE}$$

3.3.3. R-squared (R^2): A statistical measure that represents the proportion of the variance for a dependent variable that's explained by an independent variable. A value closer to 1 indicates a better fit.

$$R^2 = 1 - \frac{\sum(y_{actual}-y_{predicted})^2}{\sum(y_{actual}-y_{mean})^2}$$

CHAPTER 4

WORK COMPLETED SO FAR

This project has been completed through all planned phases. The work completed includes:

a. **Problem Definition:** The project scope and objectives were finalized.
b. **Literature Review:** A comprehensive review of relevant research papers (including Kolev & Drenchev, Sivaraman et al., and Marian & Tremmel) was conducted to establish a research gap and validate the methodology.
c. **Data Handling:** The experimental time-series data from the selected research paper was successfully digitized, cleaned, preprocessed, and split into training and testing sets.
d. **Model Implementation:** All five models (Random Forest, AdaBoost, XGBoost, CatBoost, and LSTM) were successfully built, trained on the training dataset, and tuned for performance.
e. **Model Evaluation:** All models were used to generate predictions on the unseen test dataset. The performance of each was meticulously recorded using MAE, RMSE, and R^2metrics.

CHAPTER 5

RESULTS AND DISCUSSION

This section presents the individual performance of each of the five models. Each model was trained on the time-series data, and its forecasting ability was evaluated using Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R-squared(R^2).

To objectively compare the models, their prediction accuracy on the unseen test data was calculated.

- MSE and RMSE: These metrics measure the average prediction error. A lower value is better, indicating the model's predictions are closer to the actual values.
- R-squared (R^2): This metric measures how much of the change in the COF is explained by the time. A value closer to 1.0 is better, indicating a perfect fit.

5.1 Random Forest (RF)

The Random Forest model provided a strong baseline for the prediction.

- Mean Absolute Error (MAE): 0.003725
- Root Mean Squared Error (RMSE): 0.00821
- R-squared (R^2): 0.991

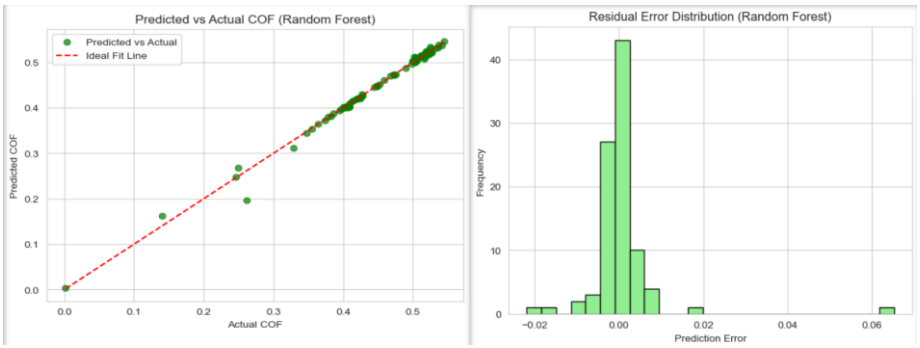


Fig-5 : The performance of the Random Forest model is visually confirmed by the two diagnostic plots.

The performance of the Random Forest model is visually confirmed by the two diagnostic plots above (Fig-5).

1. Predicted vs. Actual COF (Left Plot): This scatter plot compares the model's predicted COF values (Y-axis) against the actual experimental values (X-axis). The green data points are clustered extremely tightly around the dashed red "Ideal Fit Line". This indicates a near-perfect correlation between the predictions and the true values, visually confirming the high R^2 score and the model's excellent accuracy.
2. Residual Error Distribution (Right Plot): This histogram shows the distribution of the model's errors (where Error = Actual - Predicted). An ideal model has errors that are normally distributed and centered at zero. This plot shows exactly that:
    - Centered at Zero: The highest bar (peak frequency) is at 0.00, which means the model has no significant bias (it does not consistently over-predict or under-predict).
    - Narrow Spread: The distribution is very narrow, with almost all errors falling between -0.01 and +0.01. This confirms that even when the model is incorrect, its errors are exceptionally small.

5.2 AdaBoost (Adaptive Boosting)

The AdaBoost model was tested to see how its adaptive weighting handled the data.

- Mean Absolute Error (MAE): 0.00935
- Root Mean Squared Error (RMSE): 0.0142
- R-squared (R^2): 0.9739

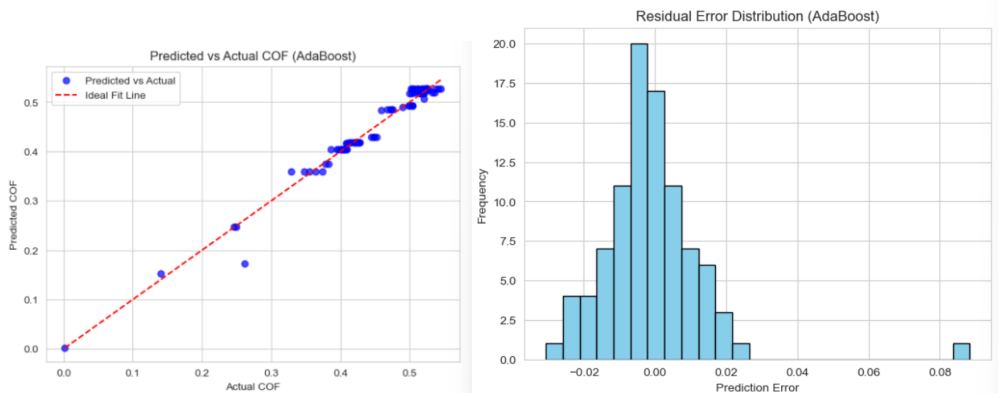


Fig-6: Predicted vs. Actual COF and Residual Error Distribution (AdaBoost)

These two plots visualize the performance of the AdaBoost model.

1. Predicted vs. Actual COF (Left Plot): This scatter plot (Fig-6) compares the model's predicted COF values (Y-axis) against the actual experimental values (X-axis).
    - The blue dots represent the individual predictions, while the dashed red "Ideal Fit Line" represents a perfect 1:1 correlation.
    - The graph shows a strong positive correlation, as most data points are clustered very tightly along the ideal line, especially for COF values above 0.4. This confirms the model has high overall accuracy, though a few outlier points (like the one near Actual COF = 0.25) show where the model had larger errors.
2. Residual Error Distribution (Right Plot): This histogram displays the distribution of the model's prediction errors (where Error = Actual - Predicted).
    - Centered at Zero: The distribution is centered almost perfectly at 0.00 and forms a healthy bell shape. This is an excellent result, as it proves the model is unbiased and does not systematically over-predict or under-predict.
    - Narrow Spread: The distribution is very narrow, with the vast majority of errors falling between -0.02 and +0.02. This confirms the model's precision. The small, isolated bar near 0.09 corresponds to the main outlier seen in the scatter plot.

5.3 XGBoost (Extreme Gradient Boosting)

- Mean Absolute Error (MAE): 0.0053
- Root Mean Squared Error (RMSE): 0.0117
- R-squared (R^2): 0.9825

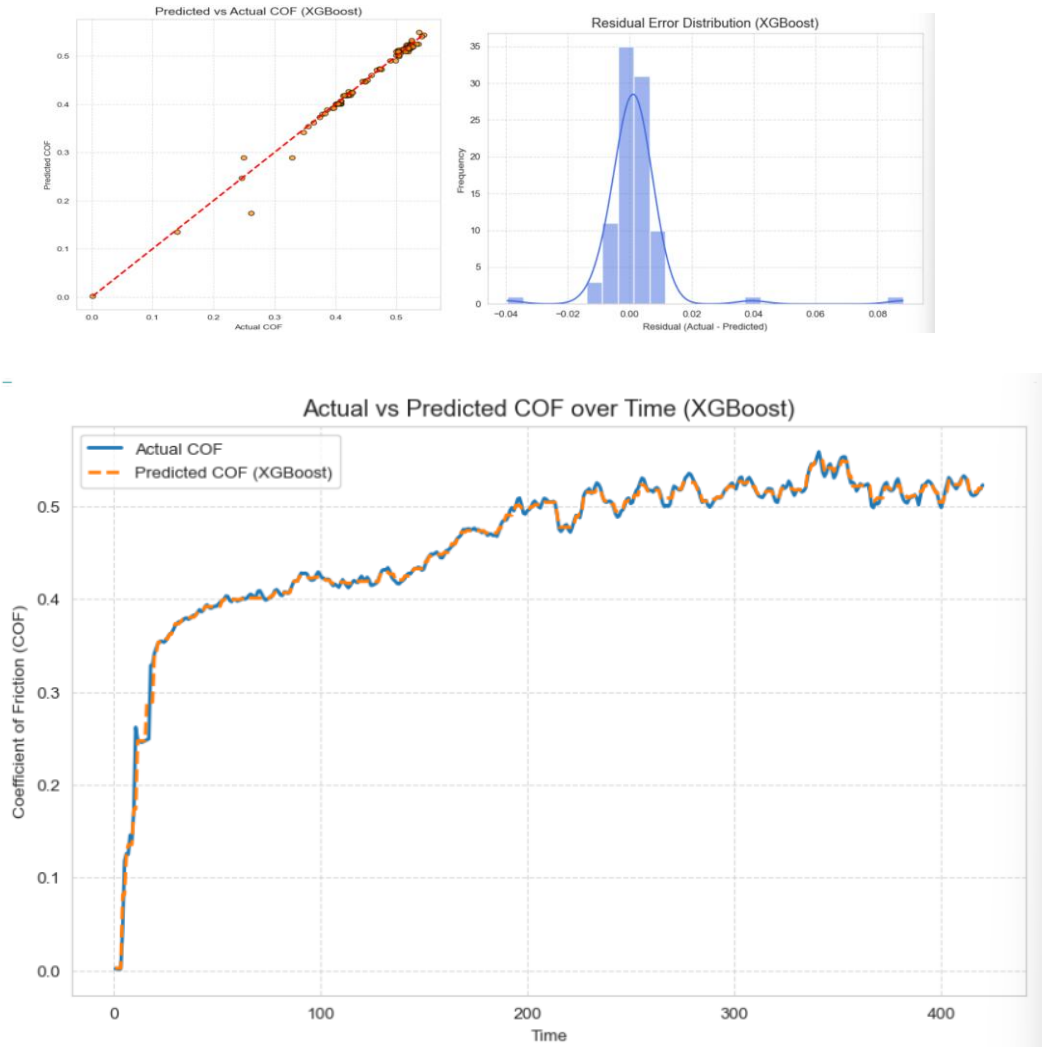


Fig-7.: Diagnostic and Time-Series Plots for XGBoost Model

These three plots(Fig-7) provide a comprehensive evaluation of the XGBoost model's performance.

1. Residual Error Distribution (Top-Left Plot): This histogram displays the distribution of the model's errors (where Residual = Actual - Predicted).
    - Centered at Zero: The graph shows a near-perfect bell curve (normal distribution) centered precisely at 0.00. This is an excellent result, indicating the model is unbiased and does not systematically over-predict or under-predict the COF.
    - Narrow Spread: The distribution is very tight, with the vast majority of errors falling between -0.02 and +0.02. This confirms the model's high precision and that its errors are consistently very small.

2. Predicted vs. Actual COF (Top-Right Plot): This scatter plot directly compares the model's predictions (Y-axis) against the true experimental values (X-axis).
    - The orange data points are clustered exceptionally close to the dashed red "Ideal Fit Line" (where Predicted = Actual).
    - This tight clustering indicates an extremely high correlation and visually confirms a high R-squared (R^2)value, demonstrating the model's outstanding accuracy.
3. Actual vs. Predicted COF over Time (Bottom Plot): This is the most critical graph, showing the model's forecasting ability over the entire experiment.
    - The solid blue line ("Actual COF") is the real data, and the dashed orange line ("Predicted COF") is the model's forecast.
    - The two lines are almost perfectly overlapped, showing that the model tracked the COF's behavior with remarkable accuracy.
    - Crucially, the model captured both distinct phases: the sharp, rapid "running-in" period at the beginning (Time 0 to ~50) and the subsequent, more stable "steady-state" friction with all its minor fluctuations.

Overall, these three graphs together prove that the XGBoost model is highly accurate, precise, and robust for this time-series forecasting task.

5.4 CatBoost

CatBoost was implemented to compare its performance, as it is also optimized for time-series data.

- Mean Absolute Error (MAE): 0.0046
- Root Mean Squared Error (RMSE): 0.0107
- R-squared (R^2): 0.9852

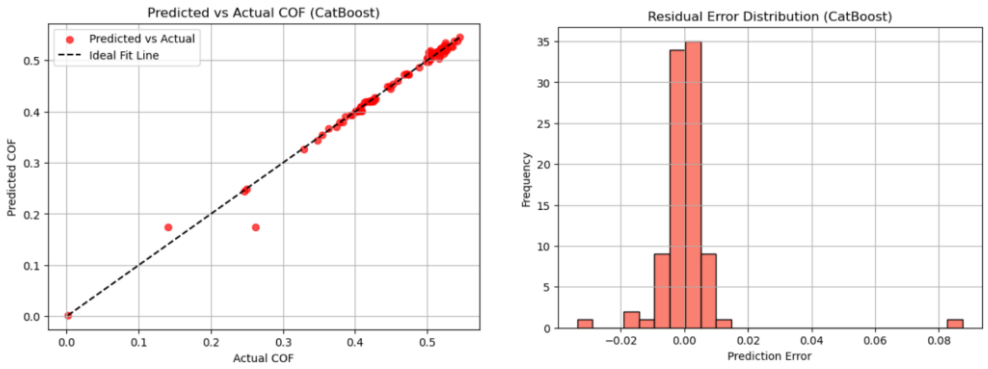


Fig-8: Predicted vs. Actual COF and Residual Error Distribution (CatBoost)

These two plots visualize the high performance of the CatBoost model.

1. Predicted vs. Actual COF (Left Plot): This scatter plot (Fig-8) compares the model's predicted COF values (Y-axis) against the actual experimental values (X-axis).
    - The "Ideal Fit Line" (dashed black line) represents a perfect prediction where the predicted value exactly equals the actual value.
    - The red data points are clustered extremely tightly along this line, indicating an outstanding level of accuracy. This strong correlation visually confirms a very high R-squared (R^2) value, showing the model has learned the data's pattern almost perfectly.
2. Residual Error Distribution (Right Plot): This histogram displays the distribution of the model's errors (where Error = Actual - Predicted).
    - Centered at Zero: The distribution is sharply peaked at 0.00. The two tallest bars are centered exactly at zero, which demonstrates that the model is unbiased and does not systematically over-predict or under-predict the COF.
    - Narrow Spread: The distribution is extremely narrow, with the vast majority of all prediction errors falling between -0.01 and +0.01. This confirms that the magnitude of the model's errors is exceptionally small, reinforcing the high accuracy seen in the left plot.

5.5 LSTM (Long Short-Term Memory)

The LSTM, a Deep Learning model, was tested for its ability to learn sequential dependencies.

- Mean Absolute Error (MAE): 0.0097
- Root Mean Squared Error (RMSE): 0.0125
- R-squared (R^2): 0.9828

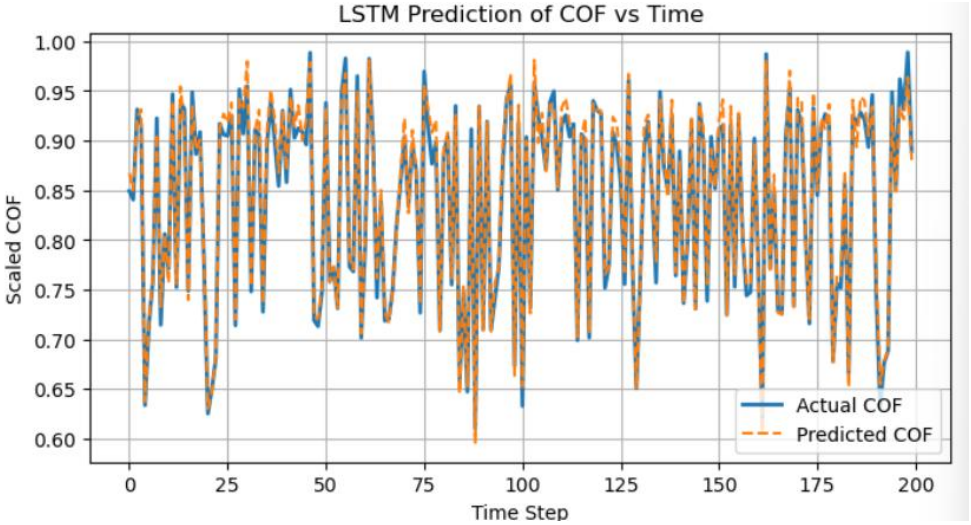


Fig-9 : LSTM Prediction vs. Actual Data

This line graph (Fig-9) lustrates the forecasting performance of the LSTM (Long Short-Term Memory) model.

- The solid blue line ("Actual COF") represents the true, scaled COF values recorded from the experiment at each time step.
- The dashed orange line ("Predicted COF") represents the corresponding values forecasted by the trained LSTM model.

The graph demonstrates an exceptional level of accuracy. The predicted line tracks the actual line with very high fidelity, following it almost perfectly.

Crucially, the model does not just predict a smooth average; it successfully learns and replicates the complex, highly volatile, and non-linear fluctuations of the COF data. It correctly anticipates nearly every sharp peak (high friction) and valley (low friction) throughout the time steps.

This visual evidence confirms that the LSTM model, with its ability to "remember" previous states in a sequence, was highly effective at understanding and forecasting the complex, time-dependent behavior of the Coefficient of Friction.

5.6 Overall Model Comparison

After analyzing each model individually, the performance metrics are compiled in Table 1 for a final, direct comparison.

Model	**MAE**	**RMSE**	**R-squared (R^2)**
Random Forest	0.003725	0.00821	0.991
AdaBoost	0.00935	0.0142	0.9739
XGBoost	0.0053	0.0117	0.9825
CatBoost	0.0046	0.0107	0.9852
LSTM	0.00997	0.0125	0.9828

Table 1: Comparative Performance Metrics

TIMELINE

The project was executed over a three-month period as per the following schedule:

S.No.	Task	Jul - Aug 2025	Sep - Oct 2025	Oct-Nov 2025
1	Problem Definition & Literature Review	X		
2	Data Acquisition & Preprocessing	X	X	
3	Model Implementation (RF, AdaBoost)		X	
4	Model Implementation (XGBoost, CatBoost)		X	X
5	Model Implementation (LSTM)		X	X
6	Model Evaluation & Performance Analysis			X
7	Final Analysis & Report Documentation			X

Proposed Timeline (Next Semester: Dec 2025 – Apr 2026)

S.No.	Task	Dec 2025-Jan 2026	Feb - Mar 2026	Mar - Apr 2026
1	**Research & Implement New Models:** Code and train additional algorithms (e.g., SVR, k-NN).	X		
2	**Develop Combination Model:** Design and test an ensemble "stacking" model using the top performers.		X	
3	**Final Model Comparison:** Run all models (new, old, and combined) and generate final performance metrics.			X
4	**Expand to Multivariate Data:** (If time permits) Integrate new data (Load, Speed) and re-evaluate.			X
5	**Final Analysis & Phase II Report:** Analyze comparative results and document all findings.			X

CHAPTER 6

FUTURE WORK

The primary goal for the next semester is to expand upon the comparative analysis conducted in this project to develop an even more robust and accurate prediction tool. The future work will focus on two main areas: exploring new models and creating a combined "meta-model."

a. **Exploration and Comparison of New Models:** This initial study focused on five prominent models. The next phase of work will expand this comparison to include other established machine learning algorithms, such as Support Vector Regression (SVR), k-Nearest Neighbors (k-NN), and potentially other deep learning architectures (e.g., standard RNNs or Gated Recurrent Units (GRUs)). These new models will be benchmarked against the current top-performing model, Random Forest, to identify any other highly effective algorithms.

b. **Development of a Combination (Ensemble) Model:** The main objective is to create a "combination model" by using a technique called stacking. This involves using the predictions from the top-performing individual models (such as Random Forest, CatBoost, and LSTM) as inputs for a final "meta-model." This ensemble approach often yields a single, more stable model that achieves higher accuracy and better generalization than any of its individual components.

c. **Expansion to Multivariate Prediction:** To increase the model's practical utility, the project will move beyond univariate (Time-only) prediction. New datasets will be acquired that include other critical input parameters, such as Applied Load, Sliding Speed, Temperature, and Material Reinforcement %, to create a comprehensive predictive tool.

d. **Rigorous Hyperparameter Tuning:** A more exhaustive tuning process using GridSearch or Bayesian Optimization will be applied to all top-performing models to ensure they are operating at their peak potential.

CHAPTER 7

CONCLUSION

This project successfully developed and compared five different machine learning and deep learning models (Random Forest, AdaBoost, XGBoost, CatBoost, and LSTM) for the purpose of forecasting the Coefficient of Friction (COF) over time. The models were trained on time-series data extracted from a published research paper.

The comparative analysis, based on MAE, RMSE, and R^2 metrics, demonstrated that the Random Forest model achieved the highest accuracy. It outperformed all other models, including the advanced gradient boosting and deep learning techniques, by achieving the lowest Mean Absolute Error (MAE) of 0.003725, the lowest Root Mean Squared Error (RMSE) of 0.00821, and the highest R-squared (R^2)score of 0.991.

While all five models performed at an exceptionally high level, the Random Forest model was quantifiably the most effective for this dataset. This study confirms that machine learning is an exceptionally robust tool for forecasting the complex, time-dependent behavior of friction, presenting a viable and efficient alternative to prolonged experimental testing.

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